

Postmortem & How To — Env, Prev Trading Day, Regression (v0.3.9)

Date: 2025-09-08

Scope: What broke, how we fixed it (without changing working logic), how we verified August, and the Git commands we used to save the checkpoint — plus what to do next.

1) Symptoms (what you saw)

- Running ``run_day_simple.py`` or ``topgappers.py`` started failing with **HTTP 401 Unauthorized** from Polygon.
- ``prev_trading_day_polygon.py`` also failed (it calls Polygon), which blocked the entire day run at the very first step.
- Previously, the same flows **worked** (we had successful runs for 2025-08-05 and 2025-09-02 earlier).

2) Root Cause (what changed)

- Your ``.env`` key was **valid** (we proved a **200** via a header auth call), but our scripts still used the older ``?apiKey=...`` **query string** style in places.
- Query string auth is **fragile** when a key has stray quotes/whitespace or when a stale machine env var shadows the project ``.env``.
- We briefly swapped the prev day helper's logic to "any 200 is a trading day," which was **wrong** on holidays (e.g. Labor Day 2025-09-01 returns 200 with empty results).

> Takeaway: the problem was auth style, not your strategy code.

3) Fixes (minimal, logic safe)

3.1 Scripts that talk to Polygon

- `scripts/topgappers.py` — **fixed**
- Load ``.env`` with `override=True`` (project ``.env`` wins).
- **Sanitize** key: `strip()` + remove accidental `` ` / `"`` quotes.
- Use HTTP **Authorization: Bearer <key>** header (no ``?apiKey=``).
- Continue to call ``prev_trading_day_polygon.py`` for prev trading day resolution.
- `scripts/prev_trading_day_polygon.py` — **fixed with logic intact**
- Keep the **original logic**: walk backward until `resultsCount > 0`` (skip weekends/holidays).
- Only change: **header auth** + sanitized key (no query param).
- Verified: ``2025-09-02 → 2025-08-29`` (Labor Day 09-01 skipped).
- `scripts/fetch_minutes_polygon.py` — **fixed**
- Same **header auth** + sanitized key; logic unchanged.
- Verified: wrote 62+ minute files for 2025-08-05 with ``.--session rth``.

3.2 What we didn't change

- No scenario logic was altered (A–E unchanged).
- No new scripts were introduced.
- We kept your runners the same (day/range/analyzers).

4) Reverification steps we ran (August D)

1) Single day sanity (D):

```
```bat
python scripts\run_day_simple.py --date 2025-08-05 --scenario D
python scripts\view_results.py --date 2025-08-05 --scenario D --preview 20 --top 5
```
```

Result: 10 trades, 4 TP (40%), TotalPnL -88.27, same symbols as before.

2) Month range (D):

```
```bat
python scripts\run_range_and_summarize.py --start 2025-08-05 --end 2025-08-31 --scenario D
python scripts\show_latest_range.py --root out\auto --scenario D
python scripts\analyze_range_explained.py --csv
out\auto\range_summary_20250805_20250831_D.csv
```
```

Summary: 79 trades, 50/29 W/L, WR 63.29%, PnL +167.57 (matches our baseline).

> These numbers match what we recalled from earlier runs, so the environment is now stable.

5) Save this checkpoint in GitHub (commands we used)

> Branch: `restore/v0.3.6` • Prior tag: `v0.3.8-2025-09-07-envfix`

1) Save the analyzer output alongside the CSV:

```
```bat
python scripts\analyze_range_explained.py --csv
out\auto\range_summary_20250805_20250831_D.csv > "Docs\versions\6. documents for
v0.3.9\AUGUST_D_range_analysis.txt"
```
```

2) Stage and commit docs + summary:

```
```bat
git add out\auto\range_summary_20250805_20250831_D.csv "Docs\versions\6. documents for
v0.3.9\AUGUST_D_range_analysis.txt"
git commit -m "Aug D baseline: WR 63.29% (+167.57). Add range summary CSV + analysis note."
git push
```
```

3) Tag the baseline for easy restore:

```
```bat
git tag -a v0.3.9-baseline-augD -m "August D baseline: WR 63.29% PnL +167.57"
```

```
git push --tags
```
```

```
> Restore later: `git checkout -b restore/v0.3.9-augD v0.3.9-baseline-augD`
```

6) What's next (simple, one at a time)

6.1 September validation (A–E), existing runners only

```
```bat
```

### D (anchor)

```
python scripts\run_range_and_summarize.py --start 2025-09-01 --end 2025-09-30 --scenario D
python scripts\analyze_range_explained.py --csv
out\auto\range_summary_20250901_20250930_D.csv
```

### E (dip)

```
python scripts\run_range_and_summarize.py --start 2025-09-01 --end 2025-09-30 --scenario E
python scripts\analyze_range_explained.py --csv
out\auto\range_summary_20250901_20250930_E.csv
```
```

6.2 (After A–E pass) Hygiene + opening RVOL gate (A/B runs)

- Hygiene: exclude warrants/units/leveraged ETFs in the universe.
- Opening RVOL gate: first 10–15m vs prior day ≥ 1.5 .

6.3 (Then) Router dryrun → enable

- One position per symbol; precedence $D \rightarrow E \rightarrow B \rightarrow A \rightarrow C$; shared risk budget.
- Dryrun first (logs only), then enable if clean.

6.4 (Later) Risk-based sizing (documented)

- Fixed \$R per trade (e.g., \$50), strict caps.
- Turn on only after September baseline is stable.

7) One line sanity checks (optional)

Prove `.env` key works (header + sanitize, read only):

```
```bat
```

```
$k = (Get-Content .\env | ? {$_. -match '^POLYGON_API_KEY='} | Select-Object -First 1) -replace
'^POLYGON_API_KEY='; $k=$k.Trim().Trim("").Trim(""); (Invoke-WebRequest -Uri
'https://api.polygon.io/v2/aggs/grouped/locale/us/market/stocks/2025-08-05?adjusted=true'
-Headers @{Authorization="Bearer $k"; 'User-Agent'='midas_v2/1.0'} -TimeoutSec 30).StatusCode
```
```

Confirm no lingering `?apiKey=` in live scripts:

```
```bat
```

```
Select-String -Path .\scripts*.py -Pattern "?apiKey=" -SimpleMatch
```

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## 8) Final note

We intentionally did not alter any scenario logic. The only changes were: ``.env`` loading with `override=True``, key sanitization, and switching to Authorization header in the three Polygon callers. The prev trading day logic remains the same and now correctly skips holidays again.